

Guan Chen

Summary

Platform architect with 13+ years designing enterprise data infrastructure across analytical, streaming, and privacy computing. As Technical Architect, defined end-to-end platform architectures, reusable abstractions, and engineering standards enabling teams to build on shared infrastructure. Operated platforms at 10B+ events/day, 1M+ connected devices, and PB-scale analytical workloads.

Experience

Senior Data Platform Engineer

Shanghai Puxin Future Internet Research Institute | Dec 2024 – Present

MIRA / ChainWeaver — Privacy Computing Platform

- Designed storage abstraction and Data Service supporting 10+ data source types, billion-scale table reads, and S3 object storage access
- Defined platform architecture: PQL → Planner → Execution DAG → Scheduler → Execution Engine → Data Service → Storage
- Established execution planning as a platform capability, replacing per-workload orchestration with declarative query submission
- Designed storage abstraction and Data Service as reusable platform contracts for execution engine and storage backend integration
- Standardized interfaces across planner, scheduler, execution engine, and data service before multi-team integration
- Built benchmark platform driving optimization with measurable evidence

Senior Data Platform Engineer / Technical Architect

Shanghai Jiayu Intelligent Robotics | Jun 2020 – Sep 2024

Connected Vehicle Telematics — National-Scale Streaming Platform (China)

- Scaled Flink-based platform from 10,000 to 1,000,000+ connected vehicles with stable production output on 1000+ vCore cluster
- Owned end-to-end platform architecture for 10B+ telemetry events/day
- Defined layer boundaries: ingestion (Kafka), computation (Flink), storage (Doris), serving (Business APIs) — each scaling independently
- Established metadata and data governance as platform primitives across national-scale telemetry datasets
- Selected Apache Doris for real-time warehouse based on columnar query performance on high-volume time-series workloads
- Enabled operational scalability via Dinky integration for Flink job lifecycle on 1000+ vCore cluster

Senior Data Platform Engineer

Ping An Securities | May 2017 – Apr 2020

Enterprise Financial Analytics — Metric Platform (China)

- Reduced 1000+ vCore batch computation from 2.5 hours to 1.5 hours via Hive tuning and Hadoop workload isolation on PB-scale storage
- Defined metric platform architecture: PB-scale Hive computation serving 100,000+ searchable Elasticsearch fields to business applications
- Solved 2000+ Hive column to 100000+ searchable field mapping through Map-based dynamic schema expansion
- Optimized 1000+ vCore analytical cluster through Hive execution tuning
- Established reusable compute-serve pattern adopted across metric onboarding

Data Platform Engineer

Shanghai Jiayin Data Technology | Restaurant Intelligence / Wi-Fi Analytics

- Built foundational streaming and batch pipelines using Storm, Hive, Hadoop

Research Engineer

Shanghai Ship and Shipping Research Institute | Scientific Computing

- Applied distributed computing to hydrodynamic simulation workloads

Skills

Architecture: Platform Design, Execution Planning, Storage Abstraction, Distributed Systems, Metadata, Data Governance

Systems: Kafka, Flink, Spark, Hive, Hadoop, Doris, Hudi, Elasticsearch, HDFS, S3

Infrastructure: Linux, Docker, Kubernetes

Languages: Java, Python, C

Education

Master, Naval Architecture and Ocean Engineering — Shanghai Ship and Shipping Research Institute

Bachelor, Hydraulic Engineering — Tianjin University

Portfolio

<https://github.com/jlty258/portfolio>
